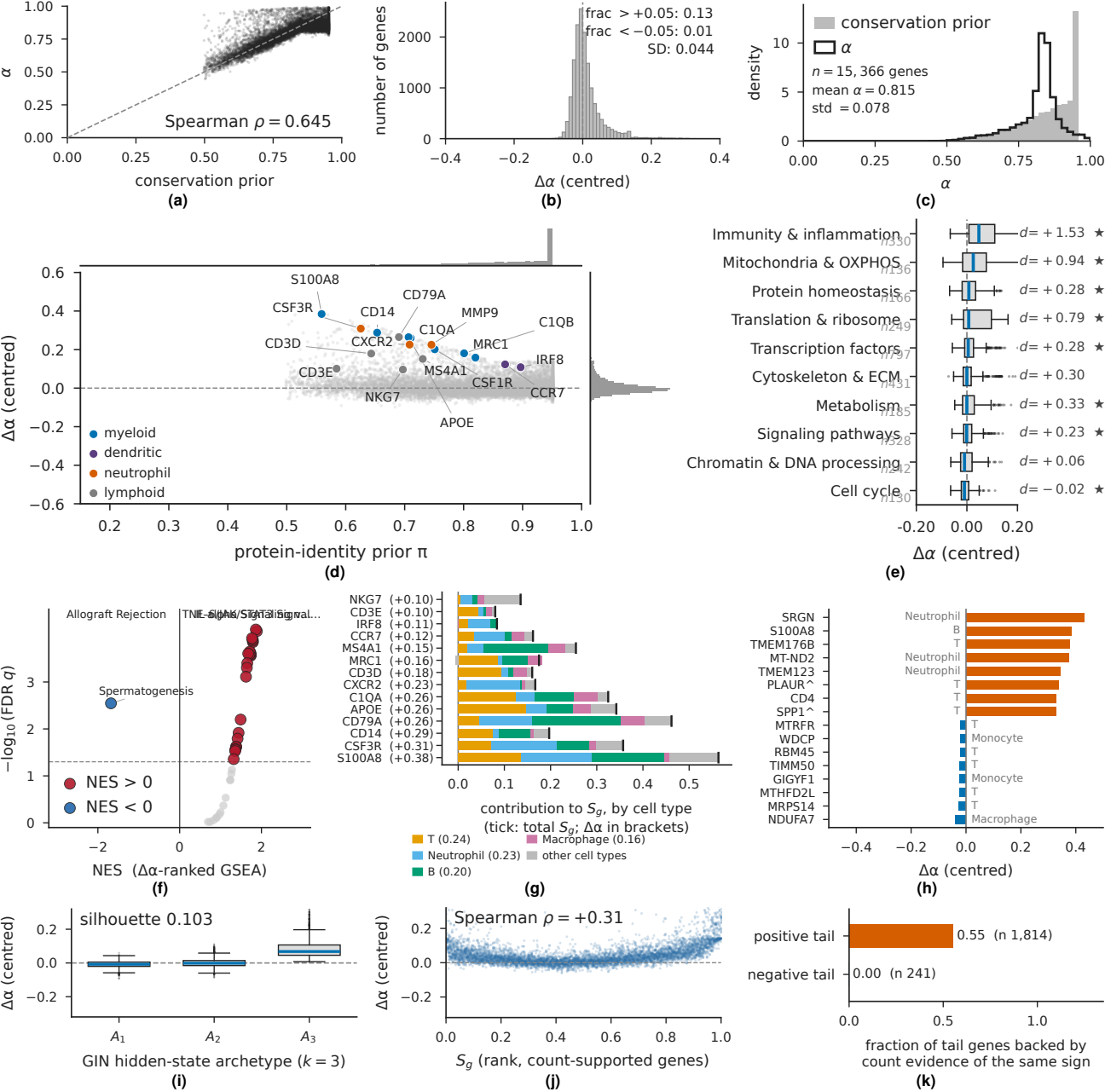

Spearman ρ = 0.645
conservation prior
α
(a)
frac > +0.05: 0.13
frac < −0.05: 0.01
SD: 0.044
number of genes
Δα (centred)
(b)
conservation prior
α
n = 15, 366 genes
mean α = 0.815
std = 0.078
density
α
(c)
S100A8
CSF3R
CD14
CD79A
MMP9
C1QA
C1QB
CD3D
CXCR2
MRC1
IRF8
CD3E
MS4A1
CSF1R
CCR7
NKG7
APOE
myeloid
dendritic
neutrophil
lymphoid
Δα (centred)
protein-identity prior π
(d)
Immunity & inflammation n=330 d= +1.53 ★
Mitochondria & OXPHOS n=136 d= +0.94 ★
Protein homeostasis n=166 d= +0.28 ★
Translation & ribosome n=249 d= +0.79 ★
Transcription factors n=97 d= +0.28 ★
Cytoskeleton & ECM n=431 d= +0.30
Metabolism n=285 d= +0.33 ★
Signaling pathways n=328 d= +0.23 ★
Chromatin & DNA processing n=42 d= +0.06
Cell cycle n=130 d= −0.02 ★
Δα (centred)
(e)
Allograft Rejection
Spermatogenesis
NES > 0
NES < 0
−log10(FDR q)
NES (Δα-ranked GSEA)
(f)
NKG7 (+0.10)
CD3E (+0.10)
IRF8 (+0.11)
CCR7 (+0.12)
MS4A1 (+0.15)
MRC1 (+0.16)
CD3D (+0.18)
CXCR2 (+0.23)
C1QA (+0.26)
APOE (+0.26)
CD79A (+0.26)
CD14 (+0.29)
CSF3R (+0.31)
S100A8 (+0.38)
contribution to S_g, by cell type
(tick: total S_g; Δα in brackets)
T (0.24)
Macrophage (0.16)
Neutrophil (0.23)
other cell types
B (0.20)
(g)
SRGN Neutrophil
S100A8 B
TMEM176B T
MT-ND2 Neutrophil
TMEM123 Neutrophil
PLAUR^ T
CD4 T
SPP1^ T
MTRFR T
WDCP Monocyte
RBM45 T
TIMM50 T
GIGYF1 Monocyte
MTHFD2L T
MRPS14 T
NDUFA7 Macrophage
Δα (centred)
(h)
silhouette 0.103
Δα (centred)
A_1
A_2
A_3
GIN hidden-state archetype ($k = 3$)
(i)
Spearman ρ = +0.31
Δα (centred)
S_g (rank, count-supported genes)
(j)
positive tail 0.55 (n 1,814)
negative tail 0.00 (n 241)
fraction of tail genes backed by count evidence of the same sign
(k)